# §5.3

# Multiplication and Division with Integers

Charged chips, number lines, patterns, and the definitions

# Today's Goal

After completing this section, you should be able to:

- Multiply two integers using the charged chip model, reading a negative first factor as removing equal groups rather than adding them
- Multiply two integers using the number line model, taking steps whose length and direction are set by the two factors
- Extend a pattern of whole number products to predict the product of two integers

# Today's Goal, Continued

- State the definition of integer multiplication and use it to evaluate a product
- Divide two integers using the charged chip model, by counting how many equal groups have to be added or removed
- State the definition of integer division and use it to evaluate a quotient

# Learning Target 13

**LT.13 Arithmetic with Integers**

I can apply multiple strategies (including the standard algorithms) for addition, subtraction, multiplication, and division of integers using appropriate representations.

# The Equal Groups Model

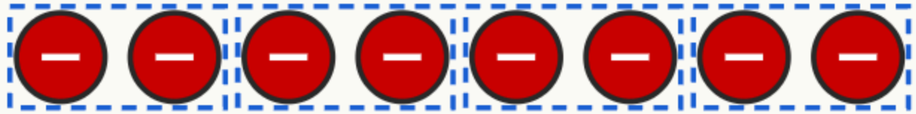
## What could $4 \times {}^{-}2$ mean?

With the charged chip model, it is helpful to think of multiplication using the equal groups model. So "$m \times n$" means $m$ equal groups of size $n$. We begin with zero and add $m$ groups of size $n$.

# Multiplying with Charged Chips

Use the charged chip model to represent and evaluate $4 \times {}^{-}2$.

$4 \times {}^{-}2$ means to add $4$ equal groups of size ${}^{-}2$. A group of size ${}^{-}2$ is represented by $2$ negative (red) chips. So we add $4$ such groups.

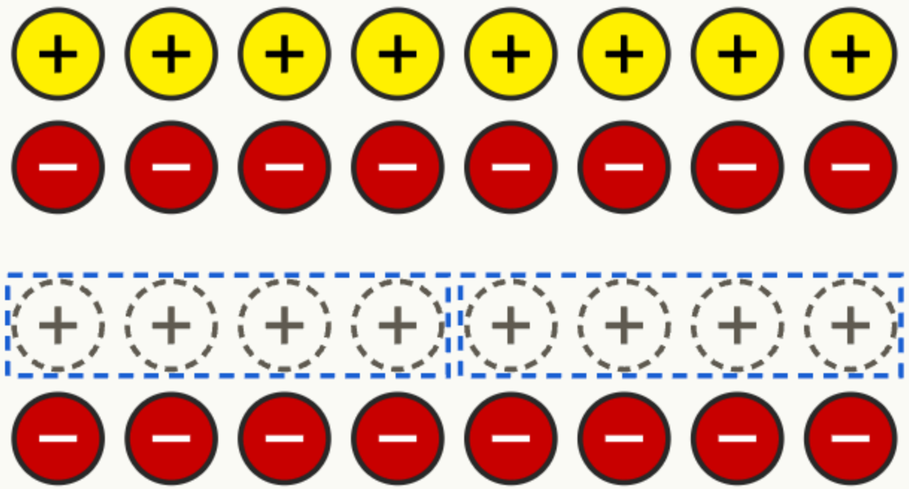


So $4 \times {}^{-}2 = {}^{-}8$.

# Removing Groups

When the number of groups is negative, we interpret that as "removing groups" rather than "adding groups". Use the charged chip model to represent and evaluate $^{-}2 \times 4$.

$^{-}2 \times 4$ means to remove $2$ equal groups of size $4$. Since we start with $0$, there is nothing to remove yet. So we must start by setting up matching pairs of positive and negative chips.



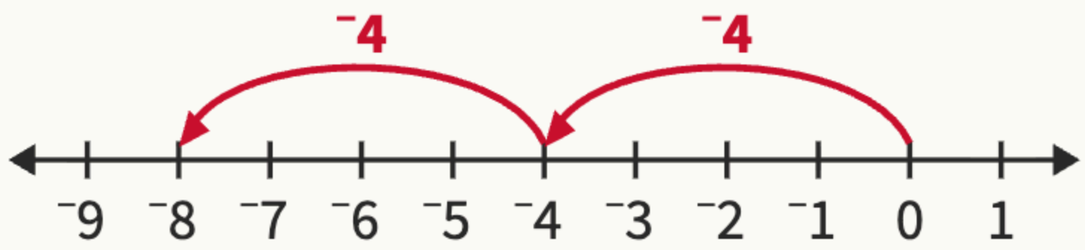
Now we can remove $2$ groups of $4$ positive chips.

So $^{-}2 \times 4 = {}^{-}8$.

# Multiplying on the Number Line

With whole numbers, we can think of $m \times n$ as meaning take $m$ steps of length $n$. When $m$ is a negative integer, we change directions. Use the number line model to represent and evaluate $^-2 \times 4$.

Since $4$ is positive, it means that each step is moving $4$ units forward. However, $^-2$ is negative. This means that forward now means moving towards the negative side. So we take $2$ steps of length $4$ to the left.




So $^-2 \times 4 = {}^-8$.

# What About $^{-}2 \times {}^{-}4$?

**Use the number line model. Which way do the steps go for $^{-}2 \times {}^{-}4$?**

Since $^{-}4$ is negative, each step is $4$ units backward.

$^{-}2$ is negative, so we turn around. Backward now means moving towards the positive side.




So we take $2$ steps of length $4$ to the right, and $^{-}2 \times {}^{-}4 = 8$.

# Find a Pattern

**Start from whole number products you know. Write a pattern that can be used to evaluate $^{-}4 \times 5$. Then write a pattern that can be used to evaluate $^{-}4 \times {}^{-}2$.**

$$1 \times 5 = 5$$
$$0 \times 5 = 0$$
$$^{-}1 \times 5 = {}^{-}5$$
$$^{-}2 \times 5 = {}^{-}10$$
$$^{-}3 \times 5 = {}^{-}15$$
$$^{-}4 \times 5 = {}^{-}20$$

$$^{-}4 \times 1 = {}^{-}4$$
$$^{-}4 \times 0 = 0$$
$$^{-}4 \times {}^{-}1 = 4$$
$$^{-}4 \times {}^{-}2 = 8$$

# Defining Multiplication with Integers

**Definition 5.3.7**

Let $m$ and $n$ be any two integers.

a. *(Multiplying Two Non-negative Integers)* If $m \geq 0$ and $n \geq 0$, then $m \times n$ is defined using whole number multiplication.

b. *(Multiplying Two Negative Integers)* If $m < 0$ and $n < 0$, then $m \times n = |m| \times |n|$.

c. *(Multiplying a Positive Integer and a Negative Integer)* If $m < 0$ and $n > 0$, then $m \times n = {}^{-}(|m| \times |n|)$.

# Do Two Negatives Make a Positive?

## Evaluate $^{-}5 \times {}^{-}2$ and $^{-}5 - {}^{-}2$.

$$^{-}5 \times {}^{-}2 = 10$$

$$^{-}5 - {}^{-}2 = {}^{-}3$$

# “Two Negatives Make a Positive”

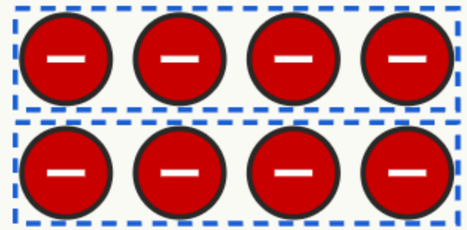
**Warning 5.3.8**

Part (b) of Definition 5.3.7 means that when we multiply two negative integers, the result will be a positive integer. To help remember this fact, students often develop the saying “two negatives make a positive”. However, when we interpret this saying literally, it is false. It depends on what you are doing with the two negatives. For example, if we multiply $^-5$ and $^-2$, we get $^-5 \times {}^-2 = 10$, which is positive. However, if we subtract, we get $^-5 - {}^-2 = {}^-3$, which is negative.

# Dividing with Charged Chips

As with multiplication, it will be useful to use the equal groups model when representing integer division with the charged chip model. In particular, $m \div n$ means "how many groups of size $n$ are in $m$?" Use the charged chip model to represent and evaluate $^-8 \div {}^-4$.

We are looking for how many groups of $^-4$ are in $^-8$. We begin by representing $^-8$ with charged chips.



Now we look for groups of $^-4$, that is, groups of $4$ negative chips.

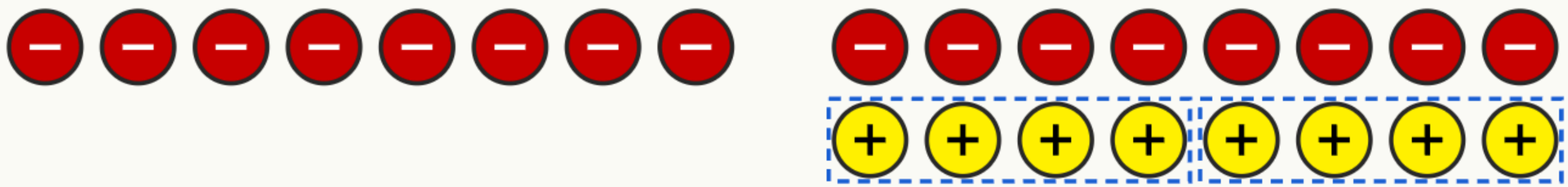
Since we can remove $2$ groups of size $^-4$, we have $^-8 \div {}^-4 = 2$.

# Adding Groups

Use the charged chip model to represent and evaluate $^{-}8 \div 4$.

**There are no positive chips. What can we do?**

It does not make sense to look for groups of $4$ to remove because there are no positive chips. Instead, we add groups of $4$ until the result is $0$.



Since we needed to "add $2$ groups of $4$", this means that $^{-}8 \div 4 = {}^{-}2$. The answer is negative because we added $2$ groups of $4$ rather than remove $2$ groups of $4$.

# Defining Division with Integers

We can define division with integers in terms of multiplication.

**Definition 5.3.11**

If $m$ and $n$ are any two integers, then $m \div n$ is the unique integer $d$ (if it exists) such that $m = n \times d$.

Note that $m \div n$ may not always be an integer. Like when dividing whole numbers, the result may not be a whole number.

# Use the Definitions

**Use Definition 5.3.7 and Definition 5.3.11 to evaluate each of the following expressions.**

(a) $^{-}7 \times {}^{-}3 = |{}^{-}7| \times |{}^{-}3| = 7 \times 3 = 21$

(b) $6 \times {}^{-}5 = {}^{-}(6 \times 5) = {}^{-}30$

(c) $^{-}24 \div 6 = {}^{-}4$, since $^{-}24 = 6 \times {}^{-}4$

(d) $15 \div {}^{-}3 = {}^{-}5$, since $15 = {}^{-}3 \times {}^{-}5$